# EE 66 Signals, Dynamics, and Information
# Fall 2026 Note 11

In the previous note, we studied inner products and norms, which let us measure lengths and angles between vectors. In this note, we introduce a new building block for working with signals: the **complex exponential** $e^{i\omega t}$. Why do we need complex numbers to study signals? Many of the signals we care about oscillate, from the voltage in a wall outlet to the pressure wave of a musical note, and complex exponentials turn out to be the most natural way to describe oscillation. We will see how they relate to sines and cosines, define frequency and period, and explore what changes when we can only observe a signal at discrete time steps.

## 11.1 A Particle on the Complex Plane

We begin with a physical example that will guide our discussion of complex exponentials.

Let $x(t) = e^{it}$ denote the instantaneous position of a particle on the complex plane. We make the following claim:

**Claim.**

1. The particle starts at position 1 (on the positive real axis) at time $t = 0$.
2. The particle stays on the unit circle at all times.
3. The particle moves counterclockwise at a constant speed of 1 radian per second.

*Proof.* First, recall that for real $\alpha$, the exponential $e^{\alpha t}$ is the unique function satisfying $\dot{x} = \alpha x$ with $x(0) = 1$. We insist on the same property when the exponent is complex: $x(t) = e^{it}$ satisfies

$$x(0) = 1, \qquad \dot{x}(t) = ix(t) \quad \text{for all } t. \tag{1}$$

We now prove each part of the claim.

**1. The particle starts at position 1.** From (1), $x(0) = 1$.

**2. The particle stays on the unit circle.** Let $x(t) = a(t) + ib(t)$ where $a$ and $b$ are real-valued functions. Then $\dot{x} = \dot{a} + i\dot{b}$. But we also know $\dot{x} = ix = i(a+ib) = -b + ia$. Matching real and imaginary parts:

$$\dot{a} = -b, \qquad \dot{b} = a. \tag{2}$$

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Multiplying the first equation by $a$ and the second by $b$, then adding:

$$a\dot{a} + b\dot{b} = -ab + ba = 0.$$

By the chain rule, the left side is $\frac{1}{2}\frac{d}{dt}(a^2 + b^2)$. Therefore:

$$\frac{d}{dt}\left(a(t)^2 + b(t)^2\right) = 0 \quad \Longrightarrow \quad |x(t)|^2 = a(t)^2 + b(t)^2 = c \quad \text{(constant for all } t\text{)}. \tag{3}$$

We know that $x(0) = 1$, so $a(0) = 1$ and $b(0) = 0$, giving $c = 1$. Therefore $|x(t)| = 1$ for all $t$: the particle is always on the unit circle.

**3. Counterclockwise at 1 radian per second.** The velocity is $\dot{x} = ix$. What does multiplying by $i$ do geometrically? Let's check a few cases:

$$i \cdot 1 = i, \qquad i \cdot i = -1, \qquad i \cdot (-1) = -i, \qquad i \cdot (-i) = 1.$$

In each case, multiplying by $i$ rotates the point 90° counterclockwise along the unit circle:

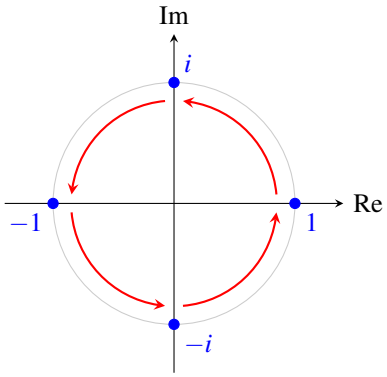



Figure 1: Multiplying by $i$ rotates each point 90° counterclockwise.

This holds for any complex number $\alpha + i\beta$:

$$i(\alpha + i\beta) = -\beta + i\alpha.$$

The point $(\alpha, \beta)$ is mapped to $(-\beta, \alpha)$, which is a 90° counterclockwise rotation about the origin.

Now let's apply this to the particle. Since $\dot{x}(t) = ix(t)$, the velocity is the position rotated 90° counterclockwise. We draw the velocity vector starting at the particle's position (Figure 2) to show where it is heading. The velocity vector is tangent to the circle and pointing counterclockwise at all times. So, the particle moves counterclockwise along the unit circle.

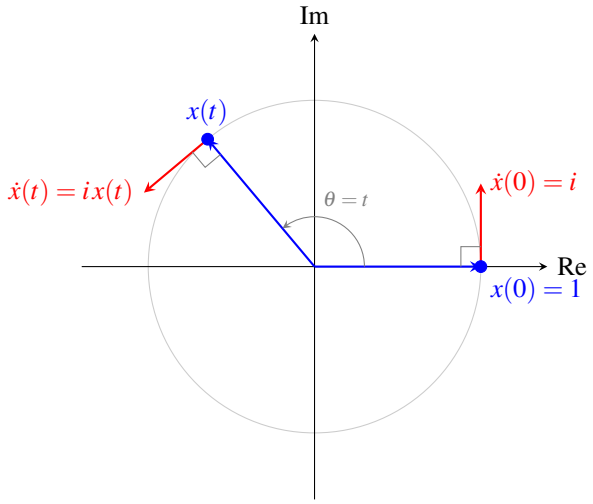



Figure 2: The velocity $\dot{x} = ix$, drawn at the particle, is tangent to the unit circle and points counterclockwise.

The speed of the particle is

$$|\dot{x}| = |ix| = |x| = \sqrt{a^2 + b^2} = \sqrt{c} = 1.$$

Therefore, the particle moves counterclockwise at a speed of 1 radian per second.

□

### 11.1.1 Euler's Formula

Now that we know the particle moves counterclockwise around the unit circle at 1 radian per second, let's derive equations for its coordinates as a function of time. After $t$ seconds the particle has swept an angle of $t$ radians. From the right triangle in Figure 3, the horizontal projection (real part) is $\cos t$ and the vertical projection (imaginary part) is $\sin t$.

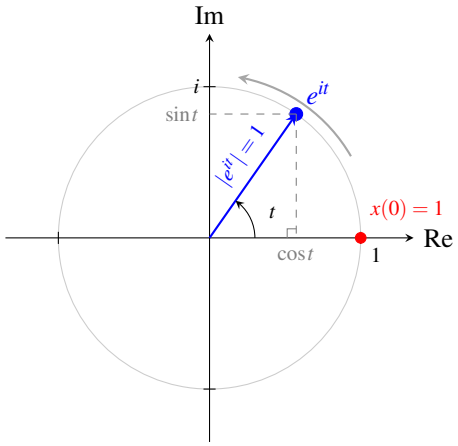



Figure 3: The particle $e^{it}$ on the unit circle. The hypotenuse has length 1, so the projections onto the real and imaginary axes are $\cos t$ and $\sin t$.

Therefore:

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta} \qquad \textbf{(Euler's formula)} \tag{4}$$

In other words, the real part of $e^{i\theta}$ is the particle's horizontal position $\cos\theta$, and the imaginary part is its vertical position $\sin\theta$.

Some useful checkpoints: $e^{i\cdot 0} = 1$, $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{i3\pi/2} = -i$, $e^{i2\pi} = 1$. So, one full revolution takes $2\pi$ seconds.

## 11.1.2 Inverse Euler's Formulas

If $e^{it}$ is a particle going counterclockwise, then $e^{-it}$ is its mirror image going clockwise at the same speed. Applying Euler's formula to both:

$$\begin{aligned} e^{it} &= \cos t + i\sin t, \\ e^{-it} &= \cos(-t) + i\sin(-t) = \cos t - i\sin t. \end{aligned}$$

Adding the two equations cancels the sine terms; subtracting cancels the cosine terms:

$$\boxed{\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}} \qquad \textbf{(Inverse Euler's)} \tag{5}$$

### 11.1.3 A Glimpse into Fourier Analysis

Look at the cosine formula again: $\cos t = \frac{1}{2}e^{it} + \frac{1}{2}e^{-it}$. This says that the cosine is a linear combination of two complex exponentials: one rotating counterclockwise at frequency $+1$ rad/s and one rotating clockwise at frequency $-1$ rad/s, each contributing with coefficient $\frac{1}{2}$.

Each complex exponential $e^{i\omega t}$ represents a single frequency. A signal like $\cos t$ is built from two such frequencies. In general, an arbitrary signal can be built from many complex exponentials at various frequencies, each with its own coefficient. Decomposing a signal in this way—finding which frequencies are present and how much each contributes—is the central theme of **Fourier analysis** and a major topic in this course.

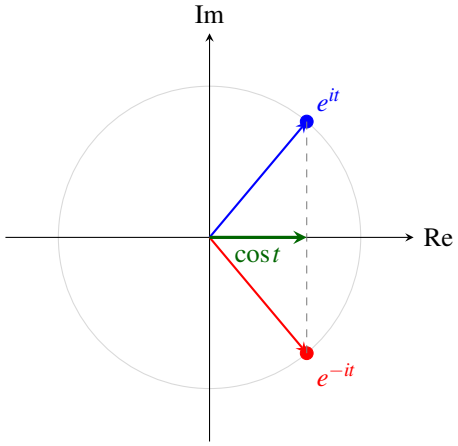



Figure 4: Adding two counter-rotating particles: $e^{it} + e^{-it}$. The imaginary parts cancel, leaving $2\cos t$ along the real axis. Dividing by 2 gives $\cos t$.

### 11.1.4 Properties of $e^{i\theta}$

The following properties all follow from Euler's formula.

$$|e^{i\theta}| = 1 \qquad \text{(on the unit circle)} \tag{6}$$

$$e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)} \qquad \text{(multiplication rotates; try multiplying out the Euler expansions)} \tag{7}$$

$$\overline{e^{i\theta}} = e^{-i\theta} \qquad \text{(reflection across real axis)} \tag{8}$$

$$e^{i\theta} \cdot e^{-i\theta} = 1 \qquad \text{(inverse property)} \tag{9}$$

$$\left(e^{i\theta}\right)^n = e^{in\theta} \qquad \text{(de Moivre; consider how this connects to (7))} \tag{10}$$

## 11.2 Frequency and Period

Before moving on, let's define some key terminology used for complex exponentials and other types of signals.

**Definition 11.1 (Period and Frequency)**:

A continuous-time (CT) signal $x(t)$ is **periodic** if there exists a $T > 0$ such that $x(t+T) = x(t)$ for all $t$.

A discrete-time (DT) signal $x[n]$ is **periodic** if there exists a positive integer $N$ such that $x[n+N] = x[n]$ for all $n \in \mathbb{Z}$.

In either case, the smallest such $T$ (or $N$) is the **fundamental period**, denoted $T_0$ (or $N_0$).

The **frequency** of a periodic CT signal describes how quickly it repeats. There are two common conventions:

- The **frequency** $f_0 = 1/T_0$, measured in **cycles per second** (Hertz, Hz).
- The **angular frequency** $\omega_0 = 2\pi/T_0 = 2\pi f_0$, measured in **radians per second** (rad/s).

The factor of $2\pi$ between the two arises because one full cycle corresponds to $2\pi$ radians. In this course we primarily use angular frequency $\omega_0$, since it appears naturally in the complex exponential $e^{i\omega_0 t}$.

In other words, a periodic signal repeats itself exactly, and its fundamental period is the length of the shortest stretch that repeats.

**Example 11.1 (Frequency of a periodic signal)**: A signal with period $T_0 = 0.5$ seconds repeats twice per second, so its frequency is $f_0 = 2$ Hz and its angular frequency is $\omega_0 = 4\pi$ rad/s.

# 11.3 Continuous-Time Complex Exponentials

Now that we have the language of frequency and period, let's generalize. The signal $e^{it}$ represents the position of a particle rotating at 1 radian per second. What if we change the speed?

## 11.3.1 Changing the Frequency: $e^{i\omega t}$

For real $\omega$, the signal $x(t) = e^{i\omega t}$ satisfies $\dot{x} = i\omega x$ with $x(0) = 1$. By the same proof as Section 11.1, the particle is still on the unit circle. What changes is the speed:

$$|\dot{x}(t)| = |i\omega|\,|x(t)| = |\omega|.$$

- $\omega > 0$: counterclockwise rotation at $\omega$ radians per second.
- $\omega < 0$: clockwise rotation at $|\omega|$ radians per second.
- $\omega = 0$: the particle sits at 1 forever (constant signal).

**Example 11.2 (Constituent frequencies of a cosine)**: What are the constituent frequencies of $\cos(5t)$?

By inverse Euler's, $\cos(5t) = \frac{1}{2}e^{i5t} + \frac{1}{2}e^{-i5t}$, so $\cos(5t)$ is made of two complex exponentials: one at $+5$ rad/s (counterclockwise) and one at $-5$ rad/s (clockwise), each with coefficient $\frac{1}{2}$.

## 11.3.2 Periodicity of CT Complex Exponentials

Consider the period of $e^{i\omega_0 t}$ with $\omega_0 \neq 0$. For periodicity, we need $e^{i\omega_0(t+T)} = e^{i\omega_0 t}$, which requires $e^{i\omega_0 T} = 1$, i.e. $\omega_0 T = 2\pi k$ for some integer $k$. The smallest positive $T$ that satisfies this is

$$T_0 = \frac{2\pi}{|\omega_0|}. \tag{11}$$

Intuitively, this is the time the particle takes to complete one full revolution.

**What about $\omega_0 = 0$?** Then $x(t) = 1$ for all $t$. Any positive shift reproduces the signal, so there is no smallest period. Constant CT signals have no fundamental period.

**Example 11.3 (Periodicity of a sum)**: Is $x(t) = e^{i3t} + e^{i\pi t}$ periodic?

The individual periods are $T_1 = 2\pi/3$ and $T_2 = 2$. For the sum to be periodic, we need a $T$ that is an integer multiple of both $T_1$ and $T_2$, which requires the ratio $T_1/T_2 = \pi/3$ to be rational. Since $\pi$ is irrational, $x(t)$ is not periodic.

## 11.3.3 From Rotation to Oscillation

So far we have pictured $e^{i\omega t}$ as a particle rotating on the unit circle. But what does this signal look like when we plot it against time? Since $e^{i\omega t}$ is complex-valued, we plot its real and imaginary parts separately:

$$\mathrm{Re}\{e^{i\omega t}\} = \cos(\omega t), \qquad \mathrm{Im}\{e^{i\omega t}\} = \sin(\omega t).$$

As the particle goes around the circle, its projections onto the real and imaginary axes trace out sinusoidal oscillations. Larger $|\omega|$ means faster rotation, which directly translates to faster oscillation.

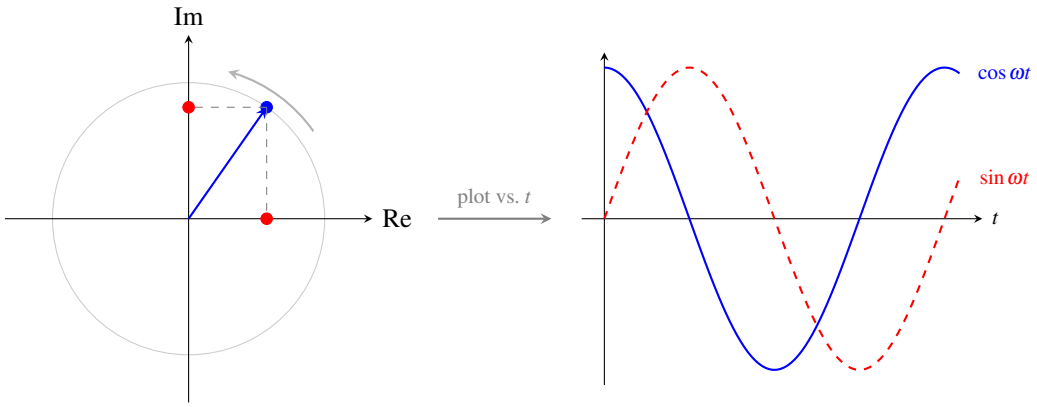



Figure 5: Left: the particle $e^{i\omega t}$ rotates on the unit circle. Right: its real part (solid) and imaginary part (dashed) traced against time are sinusoidal oscillations. Faster rotation corresponds to faster oscillation.

### 11.3.4 The General CT Complex Exponential

More generally, we can consider

$$x(t) = e^{st}, \qquad s = \sigma + i\omega, \qquad \sigma, \omega \in \mathbb{R}. \tag{12}$$

When $s$ is purely imaginary ($\sigma = 0$), we get the unit-circle rotation from before. So what role does $\sigma$ play? The magnitude of $e^{st}$ is:

$$|e^{st}| = |e^{(\sigma+i\omega)t}| = \underbrace{|e^{\sigma t}|}_{\text{amplitude}} \cdot \underbrace{|e^{i\omega t}|}_{=1} = e^{\sigma t}.$$

Since $e^{i\omega t}$ lives on the unit circle, it does not contribute to the magnitude of $e^{st}$. Its magnitude, equivalent to the amplitude of the real and imaginary oscillations, is entirely determined by $e^{\sigma t}$: $\sigma < 0$ gives a decaying oscillation, $\sigma = 0$ gives constant amplitude, and $\sigma > 0$ gives a growing oscillation. Meanwhile, $\omega$ controls the oscillation frequency exactly as before.

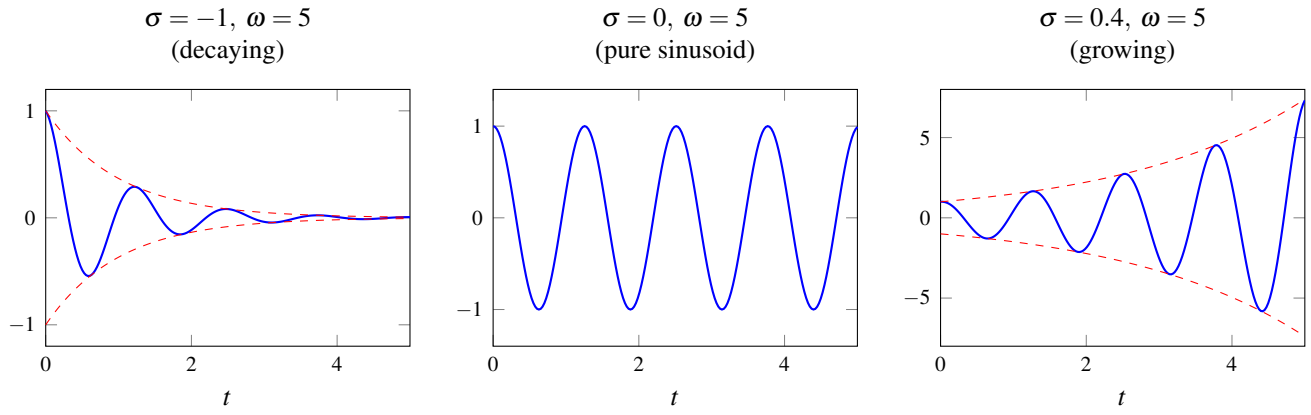



Figure 6: Re$\{e^{st}\}$ for three values of $s$ with the same $\omega$. The envelope $\pm e^{\sigma t}$ (dashed) decays when $\sigma < 0$ and grows when $\sigma > 0$.

In the next section, we will see that frequency behaves quite differently when we can only observe a signal at discrete times.

## 11.4 Discrete-Time Complex Exponentials

Now suppose we can only observe the particle at integer times $n = 0, \pm 1, \pm 2, \ldots$ This is what happens when a continuous-time signal is **sampled**: we record snapshots at regular intervals. This shift from continuous to discrete time has significant consequences for how frequency behaves.

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### 11.4.1 Definition

The **discrete-time complex exponential** with frequency $\omega_0$ is

$$x[n] = e^{i\omega_0 n}, \qquad n \in \mathbb{Z}. \tag{13}$$

Recall Euler's formula (4), which still applies in discrete time: $x[n] = \cos(\omega_0 n) + i\sin(\omega_0 n)$.

### 11.4.2 Frequencies Are $2\pi$-Periodic: Aliasing

Unlike in continuous time, distinct DT frequencies do not always produce distinct signals. Consider the DT signal at frequency $\omega_0 + 2\pi$:

$$e^{i(\omega_0+2\pi)n} = (e^{i\omega_0 n}) \underbrace{(e^{i2\pi n})}_{=1 \text{ for all } n\in\mathbb{Z}} = e^{i\omega_0 n}.$$

Therefore frequencies $\omega_0$ and $\omega_0 + 2\pi$ are identical in discrete time. In continuous time, this does not happen: $e^{i2\pi t} \neq 1$ for non-integer $t$, so different frequencies always produce different signals.

This means DT complex exponentials are **$2\pi$-periodic in frequency**. We only need to consider a single interval of length $2\pi$, such as $[0, 2\pi)$ or $(-\pi, \pi]$. Frequencies outside this range are **aliases**: they produce identical signals once sampled. Within any such interval, the two extremes are:

- $\omega_0 = 0$: $x[n] = 1$ for all $n$. Constant signal, no oscillation. This is the **slowest** DT frequency.
- $\omega_0 = \pi$: $x[n] = e^{i\pi n} = (-1)^n$, alternating $+1, -1, +1, \ldots$ This is the **fastest** possible DT oscillation.

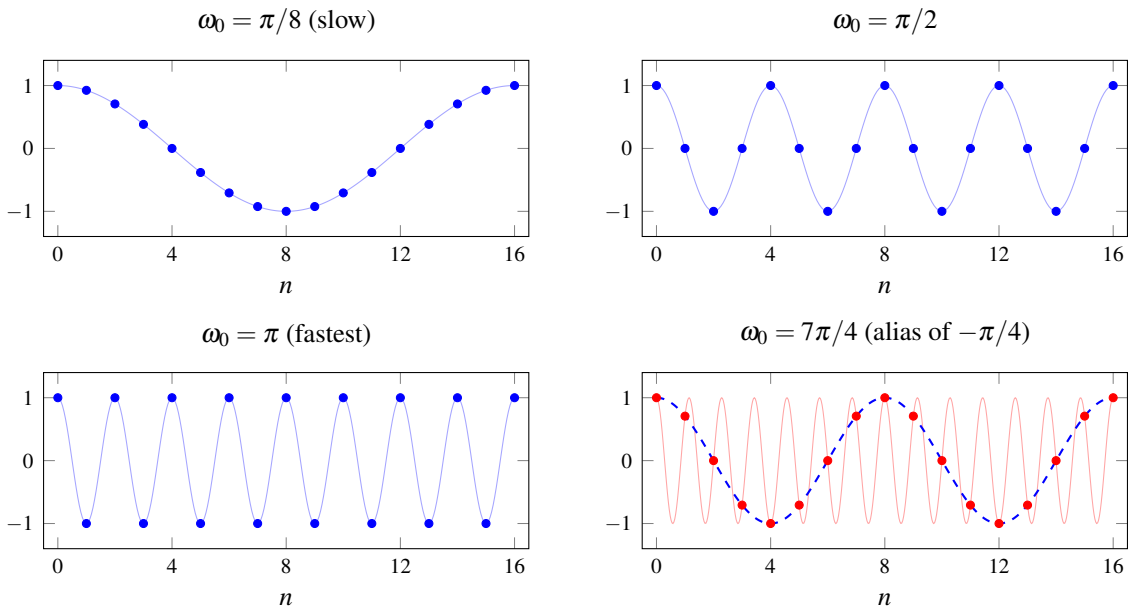



Figure 7: Each panel shows $\mathrm{Re}\{e^{i\omega_0 n}\} = \cos(\omega_0 n)$ (dots) on top of the continuous cosine $\cos(\omega_0 t)$ it samples (solid). As $\omega_0$ increases from 0 to $\pi$, the oscillation speeds up, reaching its fastest at $\omega_0 = \pi$. In the bottom right, the fast cosine at $\omega_0 = 7\pi/4$ (solid red) is **undersampled**: one sample per time step is too few to follow it, so its samples (red dots) land exactly on the slow cosine at frequency $-\pi/4$ (dashed blue), and the two frequencies are indistinguishable in discrete time.

**Physical intuition.** Think about the unit-circle particle from Section 11.1, now spinning at some very high frequency. If you only take snapshots at integer times, a fast-spinning particle can look the same as a slow-spinning one, just as a rapidly spinning wheel in a movie can appear to rotate slowly or even backward. The bottom-right panel of Figure 7 shows exactly this: a particle spinning counterclockwise at $7\pi/4$ rad per sample produces the same snapshots as one spinning *clockwise* at $\pi/4$ rad per sample, since $7\pi/4 = 2\pi - \pi/4$.

## 11.4.3 DT Periodicity

Recall from Section 11.3.2 that every CT complex exponential $e^{i\omega_0 t}$ with $\omega_0 \neq 0$ is periodic, with fundamental period $T_0 = 2\pi/|\omega_0|$. In discrete time, the story is different. The DT complex exponential $e^{i\omega_0 n}$ is periodic with period $N$ if $e^{i\omega_0(n+N)} = e^{i\omega_0 n}$ for all $n$, which requires $e^{i\omega_0 N} = 1$, i.e. $\omega_0 N = 2\pi k$ for integers $N > 0$ and $k$. Equivalently,

$$\frac{\omega_0}{2\pi} = \frac{k}{N} \in \mathbb{Q}. \tag{14}$$

In other words, a DT complex exponential is periodic exactly when its frequency is a rational multiple of $2\pi$. If $\omega_0/(2\pi)$ is irrational, **the signal is not periodic**, unlike the CT case, where $e^{i\omega_0 t}$ is always periodic for $\omega_0 \neq 0$. When $\omega_0/(2\pi)$ is rational, the fundamental period $N_0$ is the denominator of $k/N$, written in lowest terms (dividing

out any common factors between $k$ and $N$).

**Example 11.4 (Fundamental period of a DT sum)**: Find the fundamental period of $x[n] = e^{i(6\pi/7)n} + e^{i(\pi/3)n}$.

For each term, we check whether $\omega_0/(2\pi)$ is rational and find the smallest period. For the first term, $\omega_1/(2\pi) = (6\pi/7)/(2\pi) = 3/7$, which is already in lowest terms, so $N_1 = 7$. For the second term, $\omega_2/(2\pi) = (\pi/3)/(2\pi) = 1/6$, so $N_2 = 6$. Since both terms are periodic, the sum is periodic with fundamental period $N_0 = \text{lcm}(7,6) = 42$.

### 11.4.4 Converting to "Real Form"

Now that we know which DT frequencies are aliases of one another, we can use this to rewrite DT signals without any imaginary numbers. The strategy is to reduce all frequencies to $(-\pi, \pi]$, pair up conjugate frequencies, and apply the inverse Euler formulas (5).

**Example 11.5 (Converting to real form)**: Rewrite $x[n] = 5 + 3e^{i(\pi/4)n} + 2e^{i\pi n} + 3e^{i(7\pi/4)n}$ without imaginary numbers.

Since $7\pi/4 = 2\pi - \pi/4$, we have $e^{i(7\pi/4)n} = e^{-i(\pi/4)n}$. Grouping conjugate pairs:

$$\begin{aligned} x[n] &= 5 + 3e^{i(\pi/4)n} + 3e^{-i(\pi/4)n} + 2e^{i\pi n} \\ &= 5 + 3 \cdot 2\cos\left(\frac{\pi}{4}n\right) + 2(-1)^n \\ &= 5 + 6\cos\left(\frac{\pi}{4}n\right) + 2(-1)^n. \end{aligned}$$